

## HOMework

# The IF Condition

*Pseudocode practice — the alternative (if / then / else) structure*

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Solve each exercise below using pseudocode. Use only the alternative structure (if ... then ... else) — combined with read, write, and simple expressions — to solve each problem. Some exercises will require nesting an if inside another if, or combining conditions with AND, OR, and NOT.

**Reminder — syntax:** `if <condition> then <instructions1> else <instructions2>`

## Exercise 1 — Positive Number

### LEVEL 1 • BEGINNER

Read an integer *n*. If *n* is greater than 0, display the message "Positive". Use a single if ... then structure — no else branch is needed.

## Exercise 2 — Even or Odd

### LEVEL 1 • BEGINNER

Read an integer *n*. Display whether it is "Even" or "Odd", using an if ... then ... else structure.

## Exercise 3 — Maximum of Two Numbers

### LEVEL 2 • EASY

Read two numbers, *a* and *b*. Display the value of the larger one.

*Hint: make sure your algorithm still gives a correct result when *a* and *b* are equal.*

## Exercise 4 — Sign of a Number

### LEVEL 2 • EASY

Read an integer *n*. Display whether it is "Negative", "Zero", or "Positive".

*Hint: you will need to nest a second if inside the else branch of the first one.*

## Exercise 5 — Maximum of Three Numbers

### LEVEL 3 • INTERMEDIATE

Read three numbers *a*, *b*, and *c*. Display the largest of the three, using nested if structures.

## Exercise 6 — Leap Year Check

### LEVEL 4 • ADVANCED

Read a year as an integer. Display whether it is a leap year. A year is a leap year if it is divisible by 4, except for century years (divisible by 100) — those must also be divisible by 400 to be a leap year.

*Hint: try writing this two different ways: once as a single condition combining AND, OR, and NOT, and once using only nested if structures. Compare the two versions.*

## Exercise 7 — Valid Triangle & Classification

### LEVEL 5 • CHALLENGE

Read three positive numbers representing the side lengths of a potential triangle: a, b, and c.

- First, check whether the three sides can actually form a triangle — the sum of any two sides must be greater than the third side.
- If they can form a triangle, determine and display whether it is equilateral (all three sides equal), isosceles (exactly two sides equal), or scalene (all sides different).
- If they cannot form a triangle, display an appropriate message instead.

***Hint:** this one needs relational operators, logical operators, and nested if ... then ... else structures all working together.*